

Lecture 15: Review of Lectures 1 to 14

Advanced quantum mechanics WiSe 2025/2026

15.12.2025

Announcements

- Next lecture **Many body physics - electron transport** on Thursday 15.01.2026 (Thursday 17.12.2025 canceled)
- Tomorrow - exercise session
- Midterm (60 mins written) on 12.01.2026 - counts as a 0.3 points bonus

Fundamentals of quantum mechanics

The Schrödinger equation in Dirac notation

- The time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H}(\hat{p}, \hat{r}) |\psi(t)\rangle .$$

- Its formal solution involves the exponential of the Hamiltonian operator (time independent case here):

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} \hat{H}(\hat{p}, \hat{r}) t} |\psi(0)\rangle .$$

- **Step 1:** Diagonalize the Hamiltonian (stationary problem):

$$\hat{H} |E_n\rangle = E_n |E_n\rangle .$$

- **Step 2:** Expand the initial state in the energy eigenbasis:

$$|\psi(0)\rangle = \sum_n c_n(0) |E_n\rangle , \quad c_n(0) = \langle E_n | \psi(0) \rangle .$$

- **Step 3:** Evolve each component in time:

$$|\psi(t)\rangle = \sum_n c_n(0) e^{-iE_n t/\hbar} |E_n\rangle .$$

The Schrödinger equation in the position representation (one dimension)

- Introduce the position eigenstates $|x\rangle$ satisfying

$$\hat{x}|x\rangle = x|x\rangle, \quad \langle x|x'\rangle = \delta(x-x'), \quad \int dx |x\rangle \langle x| = 1.$$

- Define the **wavefunction** as

$$\psi(x, t) = \langle x|\psi(t)\rangle.$$

- Starting from

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad 1 = \int dx |x\rangle \langle x|$$

- and replacing $\hat{x} \rightarrow x$ and $\hat{p} \rightarrow -i\hbar \frac{d}{dx}$

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \hat{H}[-i\hbar \frac{d}{dx}, x] \psi(x, t).$$

- The **stationary problem** follows by separating variables $\psi(x, t) = \phi(x)e^{-iEt/\hbar}$, yielding

$$\hat{H}[-i\hbar \frac{d}{dx}, x] \phi(x) = E \phi(x),$$

i.e. the time-independent Schrödinger equation.

Example: The free particle in one dimension

- The Hamiltonian of a free particle

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

- In the position representation, solve the stationary equation

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E \psi(x).$$

- Solutions are plane waves

$$\phi_k(x) = \frac{1}{\sqrt{2\pi}} e^{ikx}, \quad E_k = \frac{\hbar^2 k^2}{2m}.$$

- Time evolution means

$$\psi(x, t) = \int_{-\infty}^{\infty} \frac{dk}{\sqrt{2\pi}} c(k) e^{ikx} e^{-i\frac{\hbar k^2}{2m} t}.$$

Example: The quantum oscillator in one dimension

The quantum harmonic oscillator in 1D

- The Hamiltonian for the quantum harmonic oscillator

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi(x) = E \psi(x),$$

- ...or use the Fock (number) basis by introducing ladder operators

$$\hat{a} = \frac{1}{\sqrt{2\hbar m\omega}}(m\omega \hat{x} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar m\omega}}(m\omega \hat{x} - i\hat{p}).$$

- Diagonalized Hamiltonian

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) = \hbar\omega \left(\hat{N} + \frac{1}{2} \right), \quad \hat{N} := \hat{a}^\dagger \hat{a}.$$

- Equidistant spectrum

$$\hat{H} |n\rangle = E_n |n\rangle, \quad E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

- The ground state $|0\rangle$ is fixed by $\hat{a}|0\rangle = 0$ and states are $|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle$. Moving up and down with

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle, \quad \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle.$$

Example: The hydrogen atom (in three dimensions)

Story of a bound electron

- Consider an electron of mass $\hat{m}$ and charge $-e$.
- Without any external potential, the system has only the kinetic energy operator

$$\frac{\hat{p}^2}{2m},$$

whose eigenstates are those of the momentum operator $\hat{p}$.

- Introducing a proton fixed at the origin provides a spherically symmetric Coulomb potential

$$V(\hat{r}) = -\frac{e^2}{4\pi\epsilon_0 \hat{r}}.$$

- Solving the Schrödinger equation yields a discrete set of bound states.
- Any additional perturbation lifts the degeneracy of these eigenstates.
- **Time-independent (degenerate) perturbation theory:** static electric and magnetic fields, spin-orbit coupling.
- **Time-dependent perturbation theory:** interaction with external laser fields; transition rates described by Fermi's golden rule.

- In the position representation the stationary Schrödinger equation is

$$\left[-\frac{\hbar^2}{2m} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right] \psi(r, \theta, \phi) = E \psi(r, \theta, \phi).$$

- It is convenient to separate the Laplacian into a radial and an angular part:

$$\nabla^2 = \underbrace{\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right)}_{\text{radial part}} + \frac{1}{r^2} \Delta_{(\theta, \phi)}.$$

- separate radial versus orbital parts and find that

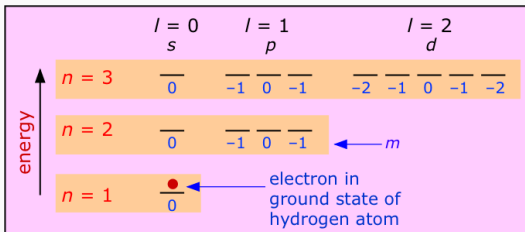
$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$$

The hydrogen atom - details and solution

- The energies of the electron in the hydrogen atom are

$$E_n = -\frac{\hbar^2}{2m a_0^2} \frac{1}{n^2}, \quad n = 1, 2, 3, \dots$$

- The energy eigenstates are n^2 degenerate



The question of perturbations

- with $\Delta E := E_e^{(0)} - E_g^{(0)}$ and

$$V_{ee} := \langle e^{(0)} | \hat{V} | e^{(0)} \rangle, \quad V_{gg} := \langle g^{(0)} | \hat{V} | g^{(0)} \rangle, \quad V_{ge} := \langle g^{(0)} | \hat{V} | e^{(0)} \rangle, \quad V_{eg} = V_{ge}^*.$$

- energy shifts are

$$E_e = E_e^{(0)} + \lambda V_{ee} + \lambda^2 \frac{|V_{ge}|^2}{\Delta E} + \mathcal{O}(\lambda^3).$$

- and the perturbed state vector is

$$|e\rangle = |e^{(0)}\rangle + \lambda |e^{(1)}\rangle + \lambda^2 |e^{(2)}\rangle + \mathcal{O}(\lambda^3),$$

- first two orders

$$|e^{(1)}\rangle = |g^{(0)}\rangle \frac{V_{ge}}{\Delta E},$$

$$|e^{(2)}\rangle = \left[-\frac{1}{2} \frac{|V_{ge}|^2}{\Delta E^2} \right] |e^{(0)}\rangle + \left[\frac{(V_{gg} - V_{ee}) V_{ge}}{\Delta E^2} \right] |g^{(0)}\rangle.$$

- Hamiltonian for spin orbit coupling

$$H_{\text{so}} = A \frac{1}{r^3} \hat{\mathbf{L}} \cdot \hat{\mathbf{S}}, \quad \text{with} \quad A \equiv \frac{Ze^2}{8\pi\epsilon_0 m_e^2 c^2}.$$

- degenerate problem - need to diagonalize Hamiltonian by introducing the total angular momentum operator

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} \otimes 1_{\text{spin}} + 1_{\text{orbital}} \otimes \hat{\mathbf{S}},$$

- diagonal form of angular part

$$\langle j, \ell, s, m_j | \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} | j, \ell, s, m_j \rangle = \frac{\hbar^2}{2} [j(j+1) - \ell(\ell+1) - s(s+1)].$$

- average of the spatial term

$$\langle r^{-3} \rangle_{n\ell} = \frac{Z^3}{a_0^3} \frac{1}{n^3 \ell(\ell + \frac{1}{2})(\ell + 1)}$$

Time dependent perturbation theory: Fermi golden rule

- Time ordered solution

$$|\psi_I(t)\rangle = |\psi_I(0)\rangle - \frac{i}{\hbar} \int_0^t dt_1 \hat{V}_I(t_1) |\psi_I(0)\rangle + \dots$$

- consider monochromatic driving

$$\hat{V}(t) = \hat{V} e^{-i\Omega t} + \hat{V}^\dagger e^{+i\Omega t},$$

- When many final states $\{|m\rangle\}$ are available, the total transition rate is the sum over all channels:

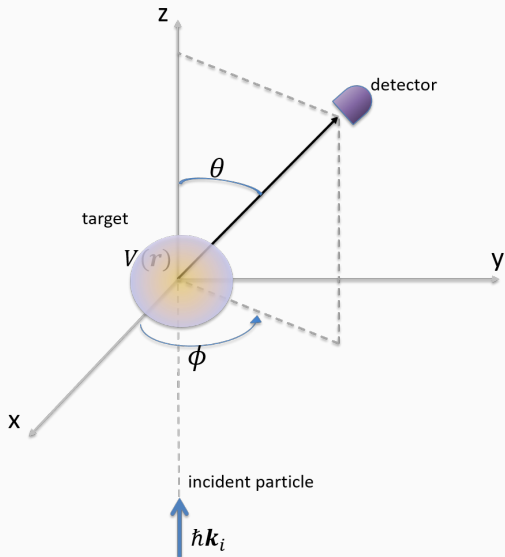
$$\Gamma_i^{(\pm)} = \sum_m \Gamma_{i \rightarrow m}^{(\pm)} = \frac{2\pi}{\hbar} \sum_m |V_{mi}|^2 \delta(E_m - E_i \mp \hbar\Omega).$$

- and in the continuum

$$\Gamma_i^{(\pm)} = \frac{2\pi}{\hbar} \int dE \rho(E) |\langle E | \hat{V} | i \rangle|^2 \delta(E - E_i \mp \hbar\Omega) = \frac{2\pi}{\hbar} \underbrace{\rho(E_i \pm \hbar\Omega)}_{\text{density of states}} |\langle E_i \pm \hbar\Omega | \hat{V} | i \rangle|^2.$$

Scattering theory

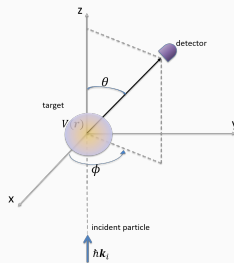
The question of scattering



The question of scattering – RELEVANCE

Scattering examples

- nuclear physics
- condensed matter
- atomic and molecular physics



Important aspects

- ratio of deBroglie wavelength versus potential range
- low energy - isotropic scattering
- high energy - many spherical components needed

The question of scattering – APPROACH

- Lippmann-Schwinger equation

$$|\psi^{(+)}\rangle = |\mathbf{k}_0\rangle + G(E_0)\hat{V} |\psi^{(+)}\rangle$$

- The Born series

$$|\psi_{\mathbf{k}_i}^{(+)}\rangle = \underbrace{|\mathbf{k}_i\rangle}_{\text{unperturbed incoming wave}} + \underbrace{\hat{G}(E_i)\hat{V} |\mathbf{k}_i\rangle}_{\text{single scattering}} + \underbrace{\hat{G}(E_i)\hat{V}\hat{G}(E_i)\hat{V} |\mathbf{k}_i\rangle}_{\text{double scattering}} + \dots$$

- Far-field: Asymptotic solution, scattering amplitude

$$\psi^{(+)}(\mathbf{r}) \simeq \frac{1}{L^{3/2}} \left[e^{i\mathbf{k}_i \cdot \mathbf{r}} + \frac{e^{ikr}}{r} f(\theta, \phi) \right].$$

- Differential and total cross section
- Partial wave decomposition
- Phase shifts: main equation

- Far-field: Asymptotic solution, scattering amplitude

$$\psi^{(+)}(\mathbf{r}) \simeq \frac{1}{L^{3/2}} \left[e^{i\mathbf{k}_i \cdot \mathbf{r}} + \frac{e^{ikr}}{r} f(\theta, \phi) \right].$$

- Differential and total cross section

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2.$$

- First Born approximation

$$f^{(1)}(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \frac{1}{(2\pi)^3} \int d^3r e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}).$$

Focus on spherically symmetric potentials

For spherically symmetric potentials, we have two very different routes:

1. **First Born approximation** — usually much faster but **approximate**:

$$f^{(1)}(\theta) = -\frac{2m}{\hbar^2} \frac{1}{q} \int_0^\infty r V(r) \sin(qr) dr, \quad q = 2k \sin \frac{\theta}{2}.$$

2. **Partial waves** — **exact**, but requires all phase shifts:

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell + 1) e^{i\delta_\ell(k)} \sin \delta_\ell(k) P_\ell(\cos \theta).$$

To compute the phase shifts, use the master formula:

$$\tan \delta_\ell(k) = \frac{kj'_\ell(ka) - L_\ell(k)j_\ell(ka)}{kn'_\ell(ka) - L_\ell(k)n_\ell(ka)}.$$

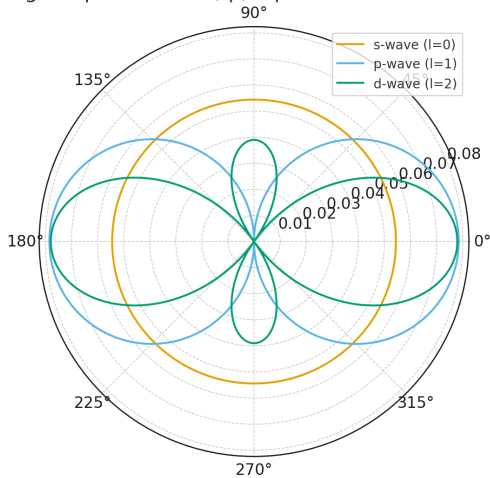
where

- a is the range of the potential
- $j_\ell(ka)$ and $n_\ell(ka)$ are known spherical Bessel and Neumann functions
- the logarithmic derivative is computed on the boundary where $V(a) \approx 0$ as

$$L_\ell(k) = \left. \frac{u_\ell^{(in)}(r)}{u_\ell^{(in)}(r)} \right|_{r=a} = \left. \frac{u_\ell^{(out)}(r)}{u_\ell^{(out)}(r)} \right|_{r=a}.$$

Partial waves scattering pattern visualized

Angular patterns of s, p, d partial waves at $kr=10$



When does the s-wave dominate?

- Low-energy condition: $kR \ll 1$.
- Only $\ell = 0$ has no centrifugal barrier

$$\frac{d^2 u_\ell}{dr^2} + \left[k^2 - \underbrace{\left(\frac{\ell(\ell+1)}{r^2} + \frac{2m}{\hbar^2} V(r) \right)}_{\text{centrifugal barrier} + \text{potential}} \right] u_\ell(r) = 0.$$

- All higher partial waves ($\ell \geq 1$) suppressed by

$$\frac{\ell(\ell+1)\hbar^2}{2mr^2}.$$

- Cross sections become isotropic.
- Full scattering determined by the scattering length a : $f(\theta) \approx -a$.

- **Ultracold atoms (μK – nK):** van der Waals range $R \sim \text{nm}$, $kR \ll 1$. Feshbach resonances, evaporative cooling $\rightarrow$ pure s-wave.
- **Electron–atom scattering below $\sim 1 \text{ eV}$:** de Broglie wavelength large; atomic size small $\rightarrow$ only $\ell = 0$ penetrates.
- **Ion–atom collisions at mK – μK :** Polarization potential; only s-wave overcomes the centrifugal barrier.
- **Cold molecular collisions:** Rotational barriers suppress $\ell > 0$; reaction rates dominated by s-wave.

- **Thermal neutrons on nuclei:** $E \sim 25 \text{ meV}$, $R \sim \text{fm} \rightarrow kR \ll 1$. Scattering essentially pure s-wave.
- **Low-energy nuclear reactions (keV):** $\ell > 0$ channels blocked by centrifugal barrier. Radiative capture and fusion rates $\rightarrow$ s-wave dominated.
- **Astrophysical plasmas (solar core):** Very low collision energies relative to nuclear size. Effective reaction rates governed almost entirely by $\ell = 0$.
- General rule: whenever de Broglie wavelength exceeds interaction range $\rightarrow$ keep only s-wave.

Many body physics — review

- Up to now: single particle Schrödinger equation

$$\hat{h} = \frac{\hat{p}^2}{2m} + V(\hat{r}), \quad \hat{h} \phi_j = E_j \phi_j.$$

- Many-body system: several particles in the same external potential

$$\hat{H} = \sum_{i=1}^N \hat{h}_i + \sum_{i < j} \hat{V}_{ij}.$$

- New key concept: **indistinguishability of identical particles.**
- Exchange of particle labels must leave probabilities invariant:

$$|\psi(r_1, r_2)|^2 = |\psi(r_2, r_1)|^2.$$

- Exchange symmetry implies

$$\psi(r_2, r_1) = \pm \psi(r_1, r_2).$$

- **Bosons** (integer spin): symmetric wavefunctions

$$\psi_B(r_1, r_2) = \begin{cases} \frac{1}{\sqrt{2}} [\phi_a(r_1)\phi_b(r_2) + \phi_a(r_2)\phi_b(r_1)], & a \neq b, \\ \phi_a(r_1)\phi_a(r_2), & a = b. \end{cases}$$

- **Fermions** (half-integer spin): antisymmetric wavefunctions

$$\psi_F(r_1, r_2) = \begin{cases} \frac{1}{\sqrt{2}} [\phi_a(r_1)\phi_b(r_2) - \phi_a(r_2)\phi_b(r_1)], & a \neq b, \\ 0, & a = b. \end{cases}$$

- Pauli exclusion principle: $\psi_F(r, r) = 0$.

Example: two particles in a harmonic trap

- Single-particle Hamiltonian

$$\hat{h} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right).$$

- Two-particle Hamiltonian

$$\hat{H} = \hat{h}_1 + \hat{h}_2.$$

- Bosonic ground state:

$$\psi_0^{(B)}(x_1, x_2) = \phi_0(x_1)\phi_0(x_2), \quad E_0^{(B)} = \hbar\omega.$$

- Fermionic ground state:

$$\psi_0^{(F)} = \frac{1}{\sqrt{2}}[\phi_0(x_1)\phi_1(x_2) - \phi_1(x_1)\phi_0(x_2)], \quad E_0^{(F)} = 2\hbar\omega.$$

Why second quantization?

- First quantization becomes cumbersome for many particles.
- Exchange symmetry must be enforced by hand.
- Interactions couple many configurations simultaneously.
- Solution: work directly with **occupation numbers**.
- Leads to the **second quantization formalism**.

- Define vacuum state

$$|0\rangle = |0_0, 0_1, 0_2, \dots\rangle.$$

- Bosonic operators

$$[\hat{a}_j, \hat{a}_k^\dagger] = \delta_{jk}.$$

- Fermionic operators

$$\{\hat{c}_j, \hat{c}_k^\dagger\} = \delta_{jk}, \quad (\hat{c}_j^\dagger)^2 = 0.$$

- Fock states

$$|n_0, n_1, \dots\rangle = \prod_j \frac{(\hat{a}_j^\dagger)^{n_j}}{\sqrt{n_j!}} |0\rangle.$$

- Single-particle contribution

$$\hat{H}_1 = \sum_j E_j \hat{a}_j^\dagger \hat{a}_j.$$

- Two-body interaction

$$\hat{H}_{\text{int}} = \frac{1}{2} \sum_{jklm} V_{jklm} \hat{a}_j^\dagger \hat{a}_k^\dagger \hat{a}_m \hat{a}_\ell.$$

- Matrix elements

$$V_{jklm} = \iint d^3r_1 d^3r_2 \phi_j^*(r_1) \phi_k^*(r_2) V(r_1 - r_2) \phi_\ell(r_1) \phi_m(r_2).$$

- Define field operators

$$\hat{\Psi}(x) = \sum_j \phi_j(x) \hat{a}_j, \quad \hat{\Psi}^\dagger(x) = \sum_j \phi_j^*(x) \hat{a}_j^\dagger.$$

- Canonical (anti)commutation relations follow automatically.
- Hamiltonian in field form

$$\hat{H} = \int dx \hat{\Psi}^\dagger(x) \hat{h}(x) \hat{\Psi}(x) + \frac{1}{2} \int dx_1 dx_2 \hat{\Psi}^\dagger(x_1) \hat{\Psi}^\dagger(x_2) V(x_1 - x_2) \hat{\Psi}(x_2) \hat{\Psi}(x_1).$$

H_2^+ molecule and the Helium atom

- System: two protons and one electron.
- Particle coordinates:

$$\mathbf{R}_1, \mathbf{R}_2 \quad (\text{protons}), \quad \mathbf{r} \quad (\text{electron}).$$

- Full wavefunction:

$$\Psi(\mathbf{R}_1, \mathbf{R}_2, \mathbf{r}).$$

- Particles are **distinguishable**.
- Protons are much heavier than the electron $\Rightarrow$ Born–Oppenheimer approximation.

- Introduce relative nuclear coordinate:

$$\mathbf{R} = \mathbf{R}_2 - \mathbf{R}_1.$$

- Factorize the wavefunction:

$$\Psi(\mathbf{R}, \mathbf{r}) = \phi(\mathbf{r}; \mathbf{R}) \chi(\mathbf{R}).$$

- First solve the electronic problem for fixed $\mathbf{R}$:

$$\hat{H}_{\text{el}}(\mathbf{R}) \phi(\mathbf{r}; \mathbf{R}) = E_{\text{el}}(\mathbf{R}) \phi(\mathbf{r}; \mathbf{R}).$$

- Then solve nuclear motion in the effective potential $E_{\text{el}}(\mathbf{R})$.

- Electronic Hamiltonian (fixed nuclei at $\pm \mathbf{R}/2$):

$$\hat{H}_{\text{el}} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 |\mathbf{r} + \mathbf{R}/2|} - \frac{e^2}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{R}/2|}.$$

- Linear combination of atomic orbitals (LCAO):

$$\phi_{\pm}(\mathbf{r}) = C_{\pm}^L \phi_{1s}(\mathbf{r} - \mathbf{R}/2) \pm C_{\pm}^R \phi_{1s}(\mathbf{r} + \mathbf{R}/2).$$

- ϕ_+ : symmetric (bonding) state, ϕ_- : antisymmetric (antibonding) state.
- Only bonding state corresponds to an equilibrium configuration.

- System: nucleus of charge $+2e$ and two electrons.
- Nuclear mass $\gg m_e \Rightarrow$ nucleus fixed at the origin.
- Electronic coordinates:

$$\mathbf{r}_1, \mathbf{r}_2, \quad r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|.$$

- Two indistinguishable fermions $\Rightarrow$ antisymmetric total wavefunction.
- First genuinely interacting many-body Coulomb problem.

- Electronic Hamiltonian:

$$\hat{H}_{\text{el}} = \hat{h} \otimes 1 + 1 \otimes \hat{h} + \hat{V}_{12}.$$

- Single-particle Hamiltonian:

$$\hat{h} = -\frac{\hbar^2}{2m_e} \nabla^2 - \frac{2e^2}{4\pi\epsilon_0 r}.$$

- Electron–electron interaction:

$$\hat{V}_{12} = \frac{e^2}{4\pi\epsilon_0 r_{12}}.$$

- Coulomb repulsion couples motions of electrons $\Rightarrow$ no exact separation of variables.

- Ground state: both electrons occupy the same spatial orbital.
- Spatial part symmetric $\Rightarrow$ spin part must be antisymmetric.
- Spin singlet (para-helium):

$$|S\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

- Total ground-state wavefunction:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \phi(\mathbf{r}_1)\phi(\mathbf{r}_2) \otimes |S\rangle.$$

- Triplet (ortho-helium) states require different spatial orbitals.

- No closed-form solution due to electron–electron interaction.
- Introduce a normalized trial orbital $\phi(\mathbf{r})$.
- Construct antisymmetrized two-electron state automatically via spin.
- Define energy functional:

$$E[\phi] = \langle \Psi | \hat{H}_{\text{el}} | \Psi \rangle.$$

- Minimize $E[\phi]$ with respect to $\phi \Rightarrow$ Hartree–Fock equation.
- Result: ground-state energy close to experimental value.

What to expect at the midterm

1. multiple choices

- 5 options, from 0 to 5 correct answers
- incorrect answer leads to minus point :(

2. exercises

- aimed to test understanding of concepts and procedure
- a separate sheet (or a few sheets) with formulas will be provided

Which of the following statements about the scattering amplitude $f(\theta, \phi)$ are correct?

- A. $|f(\theta, \phi)|^2$ gives the differential cross section
- B. $f(\theta, \phi)$ multiplies the outgoing spherical wave in the asymptotic solution
- C. $f(\theta, \phi)$ is directly measurable in an experiment
- D. $f(\theta, \phi)$ depends on the choice of quantization volume L^3
- E. $f(\theta, \phi)$ contains information about the interaction potential

Which statements about the Born–Oppenheimer treatment of the H_2^+ molecule are correct?

- A. The electronic wavefunction depends parametrically on the internuclear distance R
- B. The approximation relies on $M_{\text{nuc}} \gg m_e$
- C. Non-adiabatic couplings vanish exactly in H_2^+
- D. Proton–proton repulsion contributes to the effective nuclear potential
- E. Electron indistinguishability plays a role in this problem

exercise: two fermions, evolution, and (im)possible preparation

Two *identical fermions* in 1D are confined in a harmonic trap

$$\hat{h} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \quad E_n = \hbar\omega \left(n + \frac{1}{2} \right),$$

and the two-body Hamiltonian is $\hat{H} = \hat{h}_1 + \hat{h}_2$.

At $t = 0$ someone claims they prepared the spatial state

$$\Psi(x_1, x_2, 0) = \phi_0(x_1)\phi_0(x_2),$$

where ϕ_n is the n -th harmonic-oscillator eigenfunction.

1. Explain why this claimed preparation is (or is not) physically allowed for two identical fermions. State the correct role of spin.
2. If the two fermions are in the spin-singlet state, construct a valid *total* initial state $|\Psi(0)\rangle$ (spin $\otimes$ spatial) with both in ϕ_0 .
3. Evolve the state under $\hat{H}$: write $|\Psi(t)\rangle$ explicitly.
4. At time t , what is the probability to measure both particles in the single-particle ground level $n = 0$?

**Enjoy the break and see you on
12.01.2026!!**
